

Leading University
Department of Computer Science and Engineering
CSE – 1151 (Discrete Mathematics)
Tutorial Exam

Full Marks: 10 (5 x 2) *[Answer any 5 questions]* Time: 20 mins

1. Define Tautology and Contradiction.
2. Construct the truth table of Conjunction and Disjunction operations.
3. Show that $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent using truth tables.
4. Calculate the bitwise AND of the binary numbers 1001010 and 0101011.
5. Draw the circuit diagram: $(\neg p \vee \neg q) \wedge r$
6. Sort the following operators from lowest to highest precedence: $\wedge, \leftrightarrow, \neg, \rightarrow, \vee$

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1. Is the statement “ $x + 1 = 2$ ” a proposition? Explain the reason.
2. Construct the truth table of $p \rightarrow q$
3. Show that $\neg(p \wedge q)$ and $\neg p \vee \neg q$ are logically equivalent using truth tables.
4. Calculate the bitwise OR of the binary numbers 1011010 and 0101010.
5. Draw the circuit diagram: $\neg(p \vee q) \wedge r$
6. Define tautology with an example.

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1. Is the statement “Read this carefully” a proposition? Explain the reason.
2. Calculate the bitwise XOR of the binary numbers 1001000 and 0101111.
3. Construct the truth table for $(p \vee \neg q) \rightarrow (p \wedge q)$.
4. Draw the circuit diagram: $(p \vee q) \vee \neg r$
5. Write down De Morgan's laws of Logical Expression.
6. Construct the truth table of $p \leftrightarrow q$